← All roadmaps

Inference Engineering

Open course ↗

Follow a request from model execution to efficient production serving.

32 chapters · 226 learning units

Track your progress. Sign in or start with What Is Inference & Inference En...

FOLLOW THE PATH

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01

FOUNDATIONS & SETUP

What Is Inference Engineering?

3 units

1 What Is Inference & Inference Engineering?
NEXT ON YOUR PATH

2 The Inference Stack — Model to GPU to Production

3 Levels of Inference Engineering

02

FOUNDATIONS & SETUP

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Skip to main content

1 Math for Memory, Throughput & Speedups

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2 Vectors, Matrices & Tensor Shapes

3 Dot Products, Softmax & Attention Math

4 Probability, Sampling & Benchmark Statistics

5 Python & PyTorch for Inference

6 Transformer Architecture Refresher

7 Linux, SSH & Remote GPU Machines

8 CUDA, Drivers, PyTorch & GPU Compatibility

9 Lab — Build Your Inference Engineering Setup

03

FOUNDATIONS & SETUP

Run Your First LLM

6 units

1 Choose & Download an Open Model

2 Load, Tokenize & Inspect the Model

3 Generate & Stream Your First Tokens

4 Watch GPU Memory, Utilization & Power

5 Serve the Model Over HTTP

6 Lab — Build a Reproducible GPU Inference Server

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1 Trace the Transformer Forward Pass

2 Q, K, V & Causal Attention During Inference

3 MLP, RMSNorm & Residual Connections

4 Logits, Sampling & the Next Token

5 Build a Tiny Autoregressive Generation Loop

6 Visual Lab — Trace One Token End-to-End

05

HOW LLM INFERENCE WORKS

Prefill, Decode, KV Cache & Memory

7 units

1 Why Autoregressive Generation Recomputes Work

2 Understand & Measure the Prefill Phase

3 Understand & Measure the Decode Phase

4 Build a KV Cache

5 Calculate Model & Runtime Memory

6 Calculate KV Memory, Context & Concurrency Limits

7 Lab — Cached vs Uncached Generation & Model Fit

06

MEASUREMENT & CAPACITY

Benchmark Inference Correctly

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Skip to main content

, TPOT, ITL, E2E & Throughput

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2 Concurrency, Request Rate & Arrival Patterns

3 Short, Long, Shared-Prefix & Mixed Workloads

4 Warm-Up, Steady State & Reproducible Runs

5 P50, P95, P99 & Variance

6 Goodput, SLOs & Failure Rates

7 Benchmarking With vLLM, SGLang & GuideLLM

8 Project — Build Your Inference Benchmark Harness

07

MEASUREMENT & CAPACITY

Break One GPU

5 units

1 Establish the Single-Request Baseline

2 Sweep Concurrency & Request Rate

3 Sweep Prompt & Output Length

4 Find Saturation, SLO Failure & OOM Boundaries

5 Lab — Map the GPU Operating Envelope

08

SERVING ENGINE FUNDAMENTALS

Move to a Real Inference Engine

6 units

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Skip to main content

a vLLM Engine

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3 OpenAI-Compatible APIs & Streaming

4 Read Engine Metrics & Request State

5 Follow a Request Through vLLM

6 Lab — PyTorch vs vLLM

09

SERVING ENGINE FUNDAMENTALS

Continuous Batching & Scheduling

7 units

1 One-by-One Serving vs Static Batching

2 Iteration-Level Scheduling

3 Continuous Batching

4 Token Budgets, Max Sequences & Max Batched Tokens

5 Queueing, Fairness, Cancellation & Preemption

6 Build a Tiny Scheduler Simulator

7 Lab — Tune TTFT vs Throughput

10

SERVING ENGINE FUNDAMENTALS

PagedAttention & KV Memory Management

6 units

1 Why Contiguous KV Allocation Breaks

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Skip to main content

dAttention

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4 Allocate, Share & Reclaim KV Blocks

5 Preemption, Recomputation & KV Pressure

6 Visual Lab — Build a Paged KV Allocator

11

SERVING ENGINE FUNDAMENTALS

Optimize One GPU

4 units

1 Establish a Tunable vLLM Baseline

2 Tune GPU Memory & KV Capacity

3 Find the SLO-Safe Configuration & Headroom

4 Project — Optimize One GPU

12

SINGLE-GPU OPTIMIZATION

Prefill & Prefix Optimization

7 units

1 Reproduce the Long-Prompt Interference Problem

2 Chunked Prefill & Choosing the Chunk Size

3 Find Repeated Prefix Computation

4 Automatic Prefix Caching

5 Shared Prompts, RAG Context & Multi-Turn Reuse

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Skip to main content

— Optimize a Mixed Long-Prompt Workload

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SINGLE-GPU OPTIMIZATION

Quantization & KV Compression

8 units

1. Precision, Memory & Hardware Support
2. BF16, FP8, INT8 & 4-Bit Inference
3. AWQ, GPTQ & SmoothQuant
4. Quantized GEMM & Dequantization Overhead
5. KV Cache Quantization
6. Measure Capacity & Performance Changes
7. Measure Quality Regression
8. Project — Build the Quality/Speed/Memory Pareto Frontier

14

SINGLE-GPU OPTIMIZATION

Profiling & GPU Performance

8 units

1. Stop Guessing: Build a Profiling Workflow
2. PyTorch Profiler & Memory Snapshots
3. Nsight Systems
4. Nsight Compute
5. Profile Prefill vs Decode

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Skip to main content

7. Compute-Bound vs Memory-Bound & the Roofline Model

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8 Lab — Produce a Bottleneck Report

15

RUNTIME & KERNEL ENGINEERING

Triton & Kernel Engineering

8 units

Advanced branch. An applied serving engineer should understand the ideas, but does not necessarily need to become a kernel specialist.

1 GPU Threads, Warps, Blocks & Memory Access

2 Triton Programming Model

3 Coalescing, Tiling & Reductions

4 Write Vector & Reduction Kernels

5 Write Softmax & RMSNorm

6 Write RoPE & Fused Operators

7 Benchmark & Profile Against PyTorch

8 Challenge — Optimize a Real Inference Kernel

16

RUNTIME & KERNEL ENGINEERING

Attention Optimization

6 units

Advanced branch. An applied serving engineer should understand the ideas, but does not necessarily need to become a kernel specialist.

1 Profile Attention & Its Memory Movement

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3 FlashAttention

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4 Prefill Attention vs Decode Attention

5 Paged Attention, FlashInfer & Serving Backends

6 Lab — Compare Attention Backends

17

RUNTIME & KERNEL ENGINEERING

Compilation & CUDA Graphs

6 units

Advanced branch. An applied serving engineer should understand the ideas, but does not necessarily need to become a kernel specialist.

1 Eager Execution & Runtime Overhead

2 torch.compile & Kernel Fusion

3 Graph Breaks & Dynamic Shapes

4 CUDA Graph Capture & Replay

5 Graph Memory, Startup & Steady-State Trade-Offs

6 Lab — Compile and Capture an Inference Workload

18

RUNTIME & KERNEL ENGINEERING

Speculative Decoding

8 units

Advanced branch. An applied serving engineer should understand the ideas, but does not necessarily need to become a kernel specialist.

1 Why Autoregressive Decode Is Sequential

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3 Acceptance, Rejection & Draft Length

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4 Speculation Under Different Concurrency Levels

5 N-Gram & Prompt-Lookup Speculation

6 Medusa, EAGLE & Multi-Token Prediction

7 DFlash & Dynamic Speculation

8 Lab — Map Where Speculative Decoding Wins

19

RUNTIME & KERNEL ENGINEERING

Compare Inference Engines

6 units

Advanced branch. An applied serving engineer should understand the ideas, but does not necessarily need to become a kernel specialist.

1 Design a Fair Engine Comparison

2 vLLM vs SGLang Architecture

3 TensorRT-LLM & Hardware-Specific Optimization

4 llama.cpp & Edge-Oriented Runtimes

5 Compare Scheduling, KV, Quantization & Kernels

6 Project — Build an Engine Selection Matrix

20

MULTI-GPU & DISTRIBUTED INFERENCE

Multi-GPU & Distributed Parallelism

9 units

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Skip to main content

2 GPU Topology, PCIe, NVLink & NCCL

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3 Tensor Parallelism

4 Collectives & Communication Overhead

5 Run and Benchmark Two-GPU Inference

6 Pipeline Parallelism

7 Data Parallelism

8 Context Parallelism

9 Lab — Choose and Benchmark a Parallelism Strategy

21

MULTI-GPU & DISTRIBUTED INFERENCE

Mixture-of-Experts Inference

6 units

1 How MoE Inference Differs From Dense Models

2 Token Routing & Expert Activation

3 Expert Parallelism & Expert Placement

4 Load Imbalance & Hot Experts

5 All-to-All Communication & Grouped GEMM

6 Lab — Profile and Optimize an MoE Workload

22

MULTI-GPU & DISTRIBUTED INFERENCE

Long-Context & Tiered KV Systems

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Skip to main content

sure Long-Context Scaling

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2 Long-Context Prefill & Attention Bottlenecks

3 KV Quantization & Context-Saving Techniques

4 CPU, SSD & Remote KV Offload

5 Tiered KV Storage & Cache Movement

6 Cache Admission, Eviction & Locality

7 Lab — Push Context Beyond GPU Memory

23

MULTI-GPU & DISTRIBUTED INFERENCE

Prefill/Decode Disaggregation

7 units

1 Why Separate Prefill and Decode

2 Size Prefill & Decode Worker Pools

3 Build Prefill & Decode Workers

4 Transfer KV Between Workers

5 Measure Transfer Cost & the Crossover Point

6 Scale and Route Prefill/Decode Independently

7 Project — Reproduce a Disaggregated Serving Architecture

24

PRODUCTION INFERENCE

Production Serving & Model Lifecycle

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Skip to main content

age the Inference Service

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2 Model Weights, Caches & Cold Starts

3 Warm-Up, Compilation & Readiness

4 Health Checks, Draining & Graceful Shutdown

5 Version Models & Engine Configurations

6 Canary Releases, Shadow Traffic & Rollbacks

7 Multi-Model & Multi-Replica Serving

8 Project — Productionize Your Optimized Server

25

PRODUCTION INFERENCE

Routing, Autoscaling & Orchestration

8 units

1 Load Balance Across Replicas

2 Queue-Aware Routing

3 Prefix- & Cache-Aware Routing

4 Autoscale From Requests, Tokens & Queue Pressure

5 Warm Capacity & GPU Startup Time

6 Ray Serve LLM

7 Kubernetes, GPU Scheduling & KServe

8 Lab — Hold an SLO Under Changing Traffic

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N INFERENCE

Observability, Reliability & Security

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9 units

1. Inference Metrics, Prometheus & Grafana
2. GPU, Scheduler, KV & Cache Metrics
3. Distributed Request Tracing
4. Admission Control, Backpressure & Load Shedding
5. Timeouts, Cancellation, Retries & Priorities
6. OOM, Worker, GPU & Cache Failure Recovery
7. Chaos Testing
8. API, Network, Tenant & Model Security
9. Project — Break and Recover the Service

27

PRODUCTION INFERENCE

Inference Economics & Capacity Planning

6 units

1. Cost per Request & Cost per Million Tokens
2. Goodput per Dollar
3. Compare GPU Price/Performance
4. Capacity Planning From Benchmark Data
5. Peak Load, Failure Headroom & Idle Capacity
6. Project — Build the Cheapest System That Meets the SLO

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28

SPECIALIZATIONS & ENGINE BUILDING

Specialized Inference Workloads

10 units

1. Structured Output & Tool-Calling Inference

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4. Agentic & Multi-Turn Inference
5. Reasoning Model Inference
6. Batch Inference
7. RL Rollout Inference
8. Diffusion Inference: Denoising & Sampling
9. Diffusion Step Budgets & Quality
10. Elective Project — Choose One Specialized Workload

29

SPECIALIZATIONS & ENGINE BUILDING

Read an Inference Engine

6 units

1. Map the vLLM Codebase
2. Follow One Request End-to-End
3. Follow the Scheduler & Request State
4. Follow KV Allocation & Block Management

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6 Project — Modify and Profile One Engine Component

30

SPECIALIZATIONS & ENGINE BUILDING

Build a Mini Inference Engine

8 units

1 Build the Model Runner & Generation Loop

2 Add KV Cache

3 Add Request Batching

4 Add Continuous Batching & Scheduling

5 Add a Paged KV Allocator

6 Add Streaming & an API

7 Add Prefix Caching & Metrics

8 Project — Benchmark Your Engine Against vLLM

31

RESEARCH & CAPSTONE

Reproduce Inference Research

8 units

1 How to Read an Inference Systems Paper

2 Choose a Paper & Reconstruct the Baseline

3 Recreate the Paper Workload

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e the Implementation

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6 Run Ablations

7 Find Where the Technique Stops Working

8 Project — Compare Against Modern Engines & Publish

32

RESEARCH & CAPSTONE

Final Inference Engineering Capstone

10 units

1 Define the Workload, Quality Target & SLO

2 Choose the Model, Hardware & Engine

3 Establish a Reproducible Baseline

4 Profile and Identify the Bottleneck

5 Optimize Memory, Scheduling & Caching

6 Optimize Precision, Kernels & Runtime

7 Decide on Multi-GPU or Disaggregation

8 Add Routing, Autoscaling & Observability

9 Failure-Test and Calculate Cost & Capacity

10 Publish and Defend the Complete System

End of the course path

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